

SHIPU, China

WMO: 585690

Lat: 29.2047N

Lon: 121.9557E

Elev: 417

StdP: 14.48

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.9	33.1	11.3	9.9	34.9	15.6	12.3	36.5	25.0	43.1	22.5	43.4	10.9	340	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.0	90.7	81.0	88.7	80.3	86.9	79.7	82.5	87.9	81.5	86.5	80.5	85.3	9.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.1	164.3	85.5	80.0	158.2	84.8	79.1	153.5	83.9	46.8	88.0	45.7	86.7	44.6	85.6	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.4	22.1	19.7	27.5	95.0	3.2	2.4	25.2	96.8	23.4	98.2	21.6	99.6	19.3	101.3		
			23.3	84.3	3.4	1.3	20.9	85.2	18.9	86.0	17.0	86.7	14.6	87.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	43.8	45.5	51.2	59.5	68.0	74.2	81.9	82.3	76.5	68.7	59.2	48.7	(6)
(7)		DBStd	14.32	6.23	6.24	5.85	5.57	4.23	4.58	3.50	2.96	4.26	4.62	6.05	6.43	(7)
(8)		HDD50	527	210	151	55	3	0	0	0	0	0	7	101		(8)
(9)		HDD65	2544	657	545	430	181	18	0	0	0	19	188	506		(9)
(10)		CDD50	5417	18	26	91	287	559	727	989	1003	794	580	283	60	(10)
(11)		CDD65	1960	0	0	1	15	112	278	524	538	344	134	14	0	(11)
(12)		CDH74	15060	0	0	2	20	213	1392	5337	5611	2181	294	10	0	(12)
(13)		CDH80	4123	0	0	0	1	15	257	1654	1761	417	18	0	0	(13)
(14)	Wind	WSAvg	10.4	10.4	10.5	10.4	9.9	9.2	9.6	11.5	11.0	10.8	10.9	10.0	10.4	(14)
(15)	Precipitation	PrecAvg	57.8	2.9	3.0	4.9	4.6	5.4	8.7	4.3	7.5	7.2	3.3	3.5	2.4	(15)
(16)		PrecMax	79.5	8.0	6.0	12.5	10.6	11.0	15.7	11.3	15.4	14.4	10.4	8.7	6.9	(16)
(17)		PrecMin	31.9	0.7	0.6	1.7	1.5	2.0	0.9	0.4	1.9	1.9	0.0	0.9	0.0	(17)
(18)		PrecStd	10.2	1.6	1.5	2.6	2.0	2.6	3.4	2.7	3.6	3.5	2.7	2.1	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.3	67.4	70.4	76.7	81.9	88.6	94.1	93.6	88.8	82.2	75.5	66.5	(19)
(20)		0.4%	MCWB	59.3	60.5	60.6	66.5	72.2	80.6	81.7	81.6	80.3	75.2	68.5	61.3	(20)
(21)		2%	DB	59.0	61.2	65.6	72.9	78.4	85.7	90.9	90.9	86.4	79.3	71.8	62.5	(21)
(22)		2%	MCWB	55.2	56.6	58.2	66.3	70.8	79.1	81.1	80.9	78.8	72.6	66.9	58.5	(22)
(23)		5%	DB	55.8	57.2	62.3	70.3	76.2	83.0	88.8	89.0	84.4	77.1	69.6	60.0	(23)
(24)		5%	MCWB	52.1	53.2	57.4	65.1	70.0	77.6	80.4	80.3	77.5	70.7	65.3	55.8	(24)
(25)		10%	DB	52.6	54.1	59.4	67.7	74.2	80.3	86.9	87.1	82.4	75.2	67.2	57.6	(25)
(26)		10%	MCWB	48.8	50.4	55.4	63.3	69.1	76.1	79.9	79.8	76.3	69.1	63.0	53.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.9	61.5	63.6	70.7	75.4	81.6	83.8	84.4	82.3	78.2	71.0	63.7	(27)
(28)		0.4%	MCDWB	62.3	64.9	67.1	73.6	79.3	87.2	89.8	88.7	86.9	80.2	72.8	65.0	(28)
(29)		2%	WB	56.5	57.6	60.4	68.6	73.4	79.7	82.5	82.6	80.0	75.1	68.8	59.7	(29)
(30)		2%	MCDWB	58.3	60.7	63.6	71.0	76.1	84.4	88.4	87.5	84.2	77.4	70.9	61.5	(30)
(31)		5%	WB	52.7	54.2	58.3	66.4	72.0	78.2	81.5	81.6	78.8	72.7	66.4	56.7	(31)
(32)		5%	MCDWB	55.0	56.6	61.5	68.9	74.5	81.9	86.7	86.6	82.5	75.8	67.9	59.1	(32)
(33)		10%	WB	49.5	51.2	56.0	64.2	70.6	77.0	80.6	80.6	77.7	70.7	64.1	54.2	(33)
(34)		10%	MCDWB	51.8	53.6	58.9	66.9	73.0	79.8	85.2	85.5	81.3	73.9	66.6	57.0	(34)
(35)	Mean Daily Temperature Range		MDBR	9.3	9.5	9.9	10.3	9.5	8.0	9.1	9.0	8.8	9.5	9.5	9.8	(35)
(36)		5% DB	MCDBR	13.2	14.3	14.3	13.6	12.3	11.0	12.2	12.0	10.7	11.3	11.8	12.4	(36)
(37)			MCWBR	9.4	9.9	9.1	8.3	6.9	5.7	5.2	4.8	5.4	6.2	7.3	8.5	(37)
(38)			MCDBR	11.5	13.2	12.7	11.5	10.2	10.4	11.0	10.6	9.3	9.2	9.1	10.2	(38)
(39)		5% WB	MCWBR	9.6	10.2	9.4	8.2	7.0	5.8	5.4	5.3	5.3	6.3	7.1	8.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.550	0.592	0.634	0.645	0.611	0.617	0.577	0.582	0.558	0.538	0.536	0.531	(40)	
(41)		taud	1.772	1.711	1.657	1.690	1.815	1.852	1.976	1.949	1.975	1.920	1.868	1.823	(41)	
(42)		Ebn at Noon	208	213	216	220	228	226	235	232	233	226	211	205	(42)	
(43)		Edh at Noon	60	69	78	77	69	66	58	59	56	55	54	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	691	803	1101	1298	1396	1323	1874	1662	1309	1121	773	697	(44)	
(45)		RadStd	115	187	103	146	179	170	191	162	150	156	109	118	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page